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Sub :- Computer vision

Code :- ITA-.0501

CO2 - AT1.

Analytical problem

solving.

1) Thresholding:

i) $r \geq 128$.

output :- 255 :- 0 .

Ans:- [0, 0, 0, 255, 255, 255, 255] .

ii) b) Intensity Slicing.

values btween 90 and 170 become 255.

[25, 60, 255, 255, 255, 200, 235] .

c) percentage :-

$\frac{3}{7} \times 100$

$= 42.86\%$.

d) Mean :-

orginal = 130 .

sliced = 183.57 .

e) conclusion :- Intensity slicing highlight the required pixels.

2) Contrast Stretching :-

a)

$$m = \frac{S_2 - S_1}{r_2 - r_1}$$

$S_1 = 0.286r$.

$S_2 = S = 20 + 3(r - 70)$

$S_3 = 230 + 0.217(r - 140)$

b) output value :-

Input :- [30, 70, 90, 110, 140, 200]

output :- [9, 20, 88, 140, 230, 243] .

c) Contrast gain :-

$$\frac{230 - 20}{140 - 70}$$

$= 3.$

d) Ans :-

segments 1 and 3 compress ; segment 2 expands.

e) Conclusion :-

It improves X-ray image details.

3) Spatial filtering:

a) Mean filter.

$$\frac{699}{9} = 77.67$$

b) weighted filter.

$$\frac{1686}{16} = 105.375.$$

c) Median filter.

Middle value = 56.

d) Absolute error:-

output = −55.5

mean = 22.17

weighted = 49.875

median = 0.5.

e) Conclusion:

The median filter removes noise

best.

4)

Edge detection :-

a) Sobel :-

$G_x = 240$, $G_y = 240$.

b) Magnitude :-

$$M = \sqrt{G_x^2 + G_y^2} .$$

$$M = \sqrt{240^2 + 240^2}$$

$$= 339.41$$

Approximately $= 240 + 240 = 480$.

Error :- 41.42%.

c) Direction :-

$$\alpha = \tan^{-1}(G_y / G_x) = 45^\circ .$$

d) prewitt

$G_x = 160$, $G_y = 160$.

Magnitude $= 226.27$

Approximation $= 320$

Error $= 41.42\%$.

e) conclusion :-

Sobel :- Edge detected .

prewitt :- Edge not detected .

Threshold = 250 .

5) Global thresholding :-

a) Global mean :-

$$m_G = \frac{122}{32}$$

$$= 3.8125$$

b) iterative threshold :-

$$T = \frac{m_1 + m_2}{2}$$

$$T = \frac{1.214 + 5.833}{2}$$

$$= 3.524$$

c) otsu variance :-

→ K=2 : 5.129
→ K=3 : 5.255
→ K=4 : 5.135.

d) optimum threshold :

K=3

→ 0-3 : Background (0)
→ 4-7 : object (1)

$$18/32 \times 100$$

$$= 56.25\%.$$